

SFWRENG 3A04

Complete Lecture Notes

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Lecture 1

Introduction

- There are two types of design:
 1. Functional design.
 2. Non-functional design.
- A design principle is a rule of thumb.
 - Software architectures are critiqued according to some given design principle(s).

Lecture 2

SKIPPED

- SKIPPED

Lecture 3

SKIPPED

- SKIPPED

Lecture 4

Requirements

- Design is best defined as the process of embedding some function into a form.
- The form a design takes should be:
 1. Dependable.
 2. Sustainable.
 3. 'Harmonious' with its environment
- Functional requirements describe the functionality of the design.
 - Describes the design's reason for existence.
- Non-functional requirements describe the form that design should take.
 - Describes how the design should be manifested or instantiated
- Non-functional requirements can put specific requirements on the design's speed, security, ease-of-use, and so on.
- The environment and functional requirements of a design place constraints on its implementation.

Lecture 5

Environments

- Systems are described by their design.
 - Functional requirements provide the reason for the design's existence.
- Systems always exist in some given environment.
 - Non-functional requirements describe how the design should work with its environment.
- Systems and their environments are always interacting in some way.
 - The environment always initiates these interactions.
- A 'scenario' is a sequence of interactions between a system and its environment.
 - A global scenario is the integration of all scenarios together for some given system and environment pair.
- A 'business event' is something a system must perform when prompted by its environment.
 - Scenarios terminate when a system fulfills the business task it was prompted to perform by its environment.
- A 'business process' is the act systems engage in to fulfil business events.
 - One or more components of a system may be engaged in any given business process.
- A 'viewpoint' is a perspective that designs should take into account.
 - For example, a legal viewpoint, an economic viewpoint, etc.
 - Viewpoints are given for each business event.
 - * Inapplicable viewpoints should still be directly mentioned and specified as so.
- The modes of a system describe equivalence classes of similar system behaviours.

Lecture 6

Viewpoints

- When considering a written, non-technical description of a system:
 - Verbs often describe the system's functions.
 - Nouns often describe what the system's functions interacts with.
- The different modes of a system represent an equivalence class of the system's state.
 - A system's state describes its behaviour in some way.
 - A system's mode changes in response to business events.
- All viewpoints contribute to a system's global scenario.
- Designs should ideally be open for addition but closed for modification.

Lecture 7

Modules

I was really tired this lecture so I didn't take many notes, sorry.

- Systems are composed of modules.
 - Modules are intended to address a single concern.
 - * Modules act as an abstraction over how the system deals with a concern.
 - In this sense, modules hide information about the system.
 - A modules 'secret' is that which a module hides.
 - * A single concern may be shared across multiple business events.
 - * Modules have minimal knowledge of each other.
- There are three types of modules:
 1. Behaviour hiding modules.
 2. Decision hiding modules.
 3. Interface hiding modules.